R13

Code No: 115EH

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, September - 2021
OPERATING SYSTEMS
(Common to CSE, IT)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks

USED PAPERS SEP-2021

1.a) What are the major services of an operating system with regard to process management? Explain.
b) Give an overview of virtual machines. [8+7]

2.a) Explain the layered architecture of an operating system.
b) Explain various types of Operating System calls. [7+8]

3. Assume that the following jobs are to be executed with one processor:

Job	Burst Time	Arrival
1	10	3
2	1	4
3	2	1
4	1	2
5	5	0

Give Gantt-Chart illustrating the execution of these jobs using RR (quantum =2) and Shortest remaining time First (Preemptive). Compare their performance in terms of average turn-around time, and average waiting time ? [15]

4.a) What are the operations defined on a semaphore?
b) Discuss about Peterson solution to critical section problem. [7+8]

5.a) What are the issues in the allocation of frames? Discuss.
b) Explain in detail how demand paging is evaluated for its performance. [7+8]

6.a) Consider the page reference string 1, 0, 2, 5, 3, 0, 1, 0, 2, 4, 0, 3, 0, 4, 2, 4, 3, 4, 1, 4, 0. How many page faults occur for the Optimal and LRU Page replacement algorithms with 4 frames each?
b) Discuss about belady's anomaly. [8+7]

7.a) What is RAID? What are the different RAID levels? Explain them.
b) Explain about record blocking. [8+7]

8.a) What are the necessary conditions that lead to deadlock and define strategies for deadlock prevention.
b) Explain about implementation of language based protection. [8+7]

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